

Savitzky-Golay split command-lag search

rotor median publish period [s]: 0.00499773025513
gimbal median publish period [s]: 0.00499868392944
initial lags [s]: {'rotor': 0.004997730255126953, 'gimbal': 0.004998683929443359}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.2s gimbal=0.0382572738s cost=1.42725721099 lag_gradient={'rotor':
-1.0711882329812754, 'gimbal': -1.3552031618013771e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.014455816025556, 'finite_difference_l2_norm': 8.014450747698737,
'difference_l2_norm': 0.0008144062094443421, 'relative_error': 0.00010161715631594982},
{'direction': 'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.014457137347932, 'finite_difference_l2_norm': 8.01445206876657,
'difference_l2_norm': 0.0008144061269195979, 'relative_error': 0.0001016171292656128},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 16.824978617722085, 'finite_difference_l2_norm': 16.82497852765247,
'difference_l2_norm': 7.041562681620749e-07, 'relative_error': 4.185183732836208e-08}]
smooth width=2: rotor=0.2s gimbal=0.0384025175s cost=1.42856541773 lag_gradient={'rotor':
-1.0477739404622095, 'gimbal': -3.913954020828836e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.920400126600795, 'finite_difference_l2_norm': 8.920423608046429,
'difference_l2_norm': 0.0031809880590240804, 'relative_error': 0.00035659607646376293},
{'direction': 'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.9204030754156, 'finite_difference_l2_norm': 8.92042655724371,
'difference_l2_norm': 0.0031809880493355713, 'relative_error': 0.0003565959574828284},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 18.58421442617971, 'finite_difference_l2_norm': 18.5842131918319,
'difference_l2_norm': 4.816582088379013e-06, 'relative_error': 2.591759854855022e-07}]
smooth width=1: rotor=0.2s gimbal=0.0384548116s cost=1.43031086461 lag_gradient={'rotor':
-1.0541765715949913, 'gimbal': 0.02725619871330376} FD=[{'direction': 'negative_gradient',
'scheme': 'backward', 'step': 6.0554544523933395e-06, 'analytic_l2_norm':
10.834819107948682, 'finite_difference_l2_norm': 10.835612306079687, 'difference_l2_norm':
0.010148126028781701, 'relative_error': 0.0009365530753705309}, {'direction':
'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 10.801625367999844, 'finite_difference_l2_norm': 10.802426891003648,
'difference_l2_norm': 0.010154616537804907, 'relative_error': 0.0009400310356427181},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 23.65120343689334, 'finite_difference_l2_norm': 23.440596626876122,
'difference_l2_norm': 2.252224396742234, 'relative_error': 0.09522654716846239}]
smooth width=0.5: rotor=0.2s gimbal=0.0386694271s cost=1.43186936255 lag_gradient={'rotor':
-1.141809172692385, 'gimbal': -0.02959605263590273} FD=[{'direction': 'negative_gradient',
'scheme': 'backward', 'step': 6.0554544523933395e-06, 'analytic_l2_norm':
13.335045263661966, 'finite_difference_l2_norm': 13.33840667606788, 'difference_l2_norm':
0.0317120452520728, 'relative_error': 0.002377498753953228}, {'direction':
'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 13.343717849420043, 'finite_difference_l2_norm': 13.347095218298438,
'difference_l2_norm': 0.031734035547918725, 'relative_error': 0.0023775986481622147},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 30.838684179547126, 'finite_difference_l2_norm': 30.791410785794024,
'difference_l2_norm': 1.870558063496626, 'relative_error': 0.06065622166646202}]
strict profile iterations: 1
iteration 1: best=(0.2, 0.038669427119) cost=1.43314646154 rotor_range=(0.195002269745, 0.2)
gimbal_range=(0.03367074319, 0.043668111049) expansions=[]

selected rotor lag 0.2 s (40.0182 publish periods)
selected gimbal lag 0.038669427119 s (7.73592 publish periods)
selected objective cost 1.43314636051